

Foundation Models for Time Series Analysis: A Tutorial and Survey

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ABSTRACT

Time series analysis is a focal point within the data mining community and a cornerstone for extracting valuable insights across real-world applications. Recent advances in foundation models have fundamentally reshaped model design for time series analysis and boosted downstream tasks in practice. These approaches often leverage pre-trained or fine-tuned foundation models to harness generalized knowledge tailored for time series analysis. This survey provides a comprehensive and up-to-date overview of foundation models for time series analysis. While prior surveys have predominantly focused on either application or pipeline aspects, they often lack an in-depth understanding of mechanisms that explain why and how foundation models benefit time series analysis. To address this gap, our survey adopts a methodology-centric classification that delineates pivotal elements of time-series foundation models, including model architectures, pre-training techniques, adaptation methods, and data modalities. Overall, this survey consolidates recent advances, theoretical underpinnings, and future directions for foundation models in time series analysis.

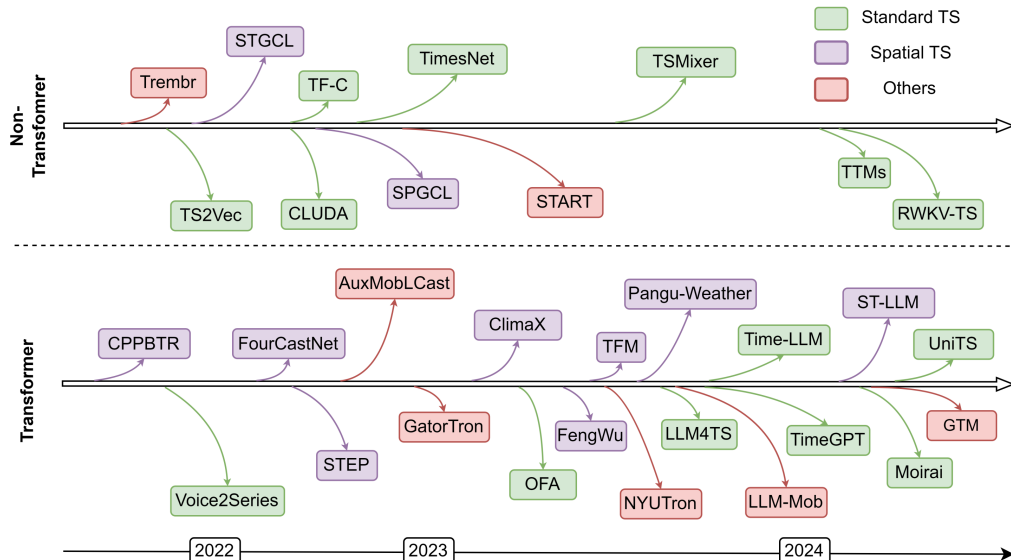


Figure 1: Roadmap of Representative Time-Series Foundation Models.